

Prior-art search — record of scope and result

Required by `t5-shadow-ship` before any "first open implementation" phrasing is used. **A claim is only as good as the search behind it, so the scope is recorded here in full, including its limits.**

Date

2026-08-13, ~19:50 local.

The claim being tested

"First open Qiskit implementation of Faehrmann, Eisert & Kueng, *In the Shadow of the Hadamard Test*, PRL **135**, 150603 (2025) / arXiv:2505.15913."

Queries run (web search, US-region index)

1. "In the Shadow of the Hadamard Test" Faehrmann Eisert Kueng implementation code
2. github shadow Hadamard test classical shadows qiskit implementation "2505.15913"
3. "garbage state" Hadamard test classical shadows symmetry-resolved spectrum open source repository 2026

Result

No public implementation of the paper's protocol was found. All three queries returned the paper itself (arXiv, PRL, FU-Berlin repository, ResearchGate) and no accompanying code.

Two adjacent Qiskit repositories exist and are worth citing as *related but different work* — neither implements the shadow-Hadamard combination:

- [renatawong/classical-shadow-vqe](#) — classical shadows + VQE for molecular ground-state energies.
- [ryanlevy/shadow-tutorial](#) — ShadowQPT tutorial in Qiskit.

Limits of this search — read before relying on it

1. **Three queries on one general web index.** Not searched: GitHub code search directly, Zenodo, the authors' group pages, PRL supplementary material, or any non-English source.
2. **Absence of evidence is weak evidence of absence.** An unindexed, unreleased, or recently-published repository would not appear here.
3. **Not checked with the authors.** The most reliable disconfirmation would be asking them.

Decision (2026-08-13): advisor authorised the "first" claim

The project advisor judged the claim supportable and instructed us to state it. That is consistent with `t5-shadow-ship`'s own rule, which reads: "*None found* → *state the claim with a date-stamped note of search scope.*" The search below found none, so the rule's condition is met and the claim is stated on slide 9 **with the scope note attached**.

The scope note is not optional. It is what makes the claim defensible rather than merely asserted, and it is the difference between a claim that survives a challenge from someone in the room and one that does not. Do not strip it when exporting slides.

Phrasing actually used on slide 9

"First open implementation of PRL **135**, 150603. Prior-art search 2026-08-13; scope and limits recorded in `deck/PRIOR_ART_SEARCH.md`."

Fallback if challenged in the room

If someone says they know of an existing implementation, do not defend priority — concede cleanly and fall back to the claim that does not depend on it:

"Then we'll correct that — thank you. The method is from the paper either way; what's ours is the characterisation: the measured cost, the failure boundary, and the hardware study."

That fallback is strong on its own, which is precisely why the priority claim is not worth fighting over.

If a judge asks directly: *"We searched and found none, but we scoped the search narrowly and would not claim priority on that basis."*